

FPT UNIVERSITY

Capstone Project Document

Lightweight Deep Learning for Robust Shrimp Disease
Detection on Mobile Devices under Diverse Environmental
Conditions

CVio	
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Definition and Acronyms

The following terms are used throughout the project reports. The same vocabulary is retained to avoid inconsistent terminology across independent submissions.

Acronym	Definition
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
AP	Average Precision
API	Application Programming Interface
ASL	Asymmetric Loss
ASL-LDAM	Custom single-label objective combining ASL-style asymmetric focusing with Label-Distribution-Aware Margin
BCE	Binary Cross-Entropy
BG	Black Gill disease
CA	Coordinate Attention
CAM	Class Activation Mapping
CBAM	Convolutional Block Attention Module
CNN	Convolutional Neural Network
CoTEGate	Project-created consensus-based residual feature-recalibration gate
CPU	Central Processing Unit
CRISP-DM	Cross-Industry Standard Process for Data Mining
CUDA	Compute Unified Device Architecture
DCFR	Project-created texture feature-recalibration component used with SimAM
DFT	Discrete Fourier Transform
DPCA	Dual-Polarity Contrast Attention
ECA	Efficient Channel Attention
EigenCAM	Principal-component-based class activation mapping method
FCN	Fully Convolutional Network
FDDEM	Frequency-Domain Detail Enhancement Module
FFT	Fast Fourier Transform
FGSM	Fast Gradient Sign Method
FPS	Frames Per Second
GAP	Global Average Pooling
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
Grad-CAM++	Generalized gradient-based class activation mapping
GRU	Gated Recurrent Unit
HiResCAM	High-Resolution Class Activation Mapping
HScore	Project-defined healthy-aware instance-segmentation decision score
HSV	Hue, Saturation, and Value color representation
IFFT	Inverse Fast Fourier Transform
IHHNV	Infectious Hypodermal and Hematopoietic Necrosis Virus
IoT	Internet of Things
IoU	Intersection over Union
JPEG	Joint Photographic Experts Group image format
KNN	K-Nearest Neighbors
LDAM	Label-Distribution-Aware Margin
LKA	Large Kernel Attention

mAP50	Mean Average Precision at an IoU threshold of 0.50
mAP50:95	Mean Average Precision over IoU thresholds 0.50 to 0.95
NMS	Non-Maximum Suppression
PAN-FPN	Path Aggregation Network and Feature Pyramid Network
RAM	Random Access Memory
ReLU	Rectified Linear Unit
RGB	Red, Green, and Blue color representation
ROI	Region of Interest
SimAM	Simple, parameter-free attention module based on neuron energy
TFLite	TensorFlow Lite
TIMM	PyTorch Image Models library
UAT	User Acceptance Testing
UI/UX	User Interface and User Experience
WBS	Work Breakdown Structure
WSSV	White Spot Syndrome Virus
WSSV_BG	Four-class image-level co-infection category containing BG and WSSV evidence
XAI	Explainable Artificial Intelligence
YHV	Yellow Head Virus
YOLO	You Only Look Once model family

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